

IMPACT OF EMPLOYEE AND CAPITAL ON MICROFINANCE PERFORMANCE: A CASE STUDY OF BANGLADESH

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Abstract

Capital investment to reduce the operating cost and gain financial sustainability is a common feature of formal banking institutions. However, in the case of informal lending institutions like microfinance, allocation of resources to attain financial sustainability while at the same time providing financial services to the poor community is an interesting area for exploration. Thus, this study endeavors to establish an empirical relationship of microfinance resources (labor and capital) with its quality (depth), quantity (breadth) and financial sustainability in Bangladesh. We used the system-generalized methods of moment (GMM) with a panel dataset of 176 microfinance institutions (MFIs) between 2007 and 2012. The study finds that higher capital asset ratio (as a proxy of real capital) and number of employees had negative and positive impacts towards quality, quantity and financial sustainability of microfinance institutions respectively. Furthermore, we have also estimated that higher capital asset ratio increases the MFIs administrative cost. These findings produce strong policy guidelines for the microfinance sector in Bangladesh, particularly for management of the MFIs. Individual MFIs should take precautionary measures before they shift their resources, mostly capital goods, by contrasting the use of labor in their operation.

Key Words: employee, capital, microfinance institutions, sustainability, Bangladesh

Introduction

Modern microfinance institutions (MFIs) have two dual missions - social outreach and financial sustainability - also known as ‘double bottom line’ (Ben Soltane, 2014; Conning, 1999; Hermes, Lensink, & Meesters, 2011; Louis, Seret, & Baesens, 2013; Piot-Lepetit & Nzongang, 2014; Yaron, 1994). Social outreach (sometimes known simply as ‘outreach’) is defined as providing financial services (e.g. credit, savings, insurance etc.) to poor communities, while financial sustainability means the ability to generate enough revenue to cover total cost. According to Conning (1999) the two components of social outreach consist of ‘depth’ and ‘breadth’, implying

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providing credit to the poorest of the poor and extending financial facilities to the wider clients respectively. Quayes (2012), termed these depth and breadth of outreach as ‘quality’ and ‘quantity’ of microfinance. MFIs are also known as ‘Hybrid Organization’ as they are a combination of banking (profitability) and development (poverty alleviation) (Tchuigoua, 2014).

Despite the labor²-intensive nature of microfinance, the sector has recorded substantial investment in capital goods to lower the operational costs and to achieve financial sustainability (Ganga & Rizavi, 2006; Guntz, 2011; Nourse, 2001). Scholars have argued that as a result of giving more focus to financial sustainability, the original mission of MFIs, that of social outreach, is being de-emphasized in many cases (Louis et al., 2013). Focusing more on financial sustainability needs and de-emphasizing social outreach needs is termed as ‘mission drift’ - (Annim, 2012; Armendáriz & Szafarz, 2011; Augsburg & Fouillet, 2013; Conning, 1999; Copestake, 2007; Hermes et al., 2011; Woller, 2007). Researchers observed that there is no or neutral correlation between financial sustainability and social outreach (Ben Soltane, 2012; Louis et al., 2013; Mersland & Strøm, 2010; Piot-Lepetit & Nzongang, 2014; Zerai & Rani, 2012). Furthermore, a complementary relationship exists between these two objectives as they are interrelated. (Adhikary & Papachristou, 2014; Quayes, 2012). Whether the increased focus on financial sustainability through positive shift of capital investment (physical asset, capital equipment, etc. finance by capital) deteriorated the social outreach performance are still lacking in the existing literatures.

Although previous studies have empirically tried to establish the relationship between these two dimensions, researchers have barely scratched the surface when determining the causes of such outcome. However, with few recent exception, Serrano-Cinca and Gutiérrez-Nieto (2014) have used long tail theory to explain the mission drift and categorized the MFIs in terms of ‘mission-centered’ or ‘otherwise’. Explaining why such trade-off may happen or not, is still not sufficient in the academic literature. More specifically, whether more investment in capital or labor causes better financial sustainability and less social outreach or *vice versa*, needs to be addressed. Hence, this study endeavors to explain the relationship between financial sustainability and social outreach (quality and quantity) by looking into the resources, particularly labor and capital that microfinance institutions use to generate their services.

This study is expected to contribute in several ways to the existing literature. Firstly, the findings are significant to management or higher authorities of the institutions. This study will examine how the relationship between labor and capital may provide financial sustainability and social outreach and how the variation of MFIs resource allocation effects the performance of both. Secondly, this study conceptualizes both objectives of microfinance based on resource allocation that will add to the growing debate of financial sustainability versus social outreach mechanisms. Furthermore, output elasticity and economies of scale are estimated that can provide managerial

² Labor and Employee are interchangeably used throughout the article.

guidelines to the authorities for upgrading their operation. Further policy implications are discussed based on the empirical estimation.

The organization of this study is as follows. We start the discussion by analyzing the recent literature. This is followed by deriving the theoretical framework and hypothesis. Next, estimation, variables and data are provided in the methodology section. Empirical findings are discussed in section four. This study concludes by providing some policy implication and areas for future research.

2. Literature Review

Microfinance Institutions aimed to cater to the poor by providing financial services because the poor are largely excluded from the mainstream financial sector (Morduch, 2000). However, to provide a healthy flow of credit and sustainable development of the sector, MFIs also need to be financially sustainable (Serrano-Cinca & Gutiérrez-Nieto, 2014; Zerai & Rani, 2012). There is a hefty debate among ‘welfarists’ and ‘institutionalists’ regarding the two objectives of MFIs. On the one hand, ‘welfarists’ argued that the objective of microfinance institutions creation should be largely focused on catering to the poorest of the poor. On the other hand, ‘institutionalists’ argue that financial viability for future growth of the sector will lead to a greater impact for the poor community (Ben Soltane, 2012; Hermes et al., 2011). However, meeting these two conflicting objectives concurrently is still a challenge that is highly debated, and may be incompatible or complementary, depending on initial assumptions (Gutiérrez-Nieto, Serrano-Cinca, & Mar Molinero, 2007; Woller & Schreiner, 2002). Given such outcomes, resource allocation strategy is very significant to the microfinance operation as it will provide better synergies and adjust the gap between the dual objectives by reallocating their resources.

The main reason behind the contrasting relationship between financial and social outreach are basically the outcome of profit maximization, which inhibits or curtails the social mission (Cull, Demirgüç-Kunt, & Morduch, 2011). As adoption of technology incurs less operational cost generally (this may be or may be not true for microfinance) and increases the profit, it is the more chosen path. There is scant research that discusses how more expenditure on capital may curtail social outreach. Most recently, Serrano-Cinca and Gutiérrez-Nieto (2014) have used long tail theory to explain the ‘mission-drift’ and categorized the MFIs in terms of ‘mission-centered’ or ‘otherwise’. They have found that those MFIs who stick with their social outreach mission are able to occupy a niche market at the end of the long tail without observing mission-drift.

Regarding the characteristics of MFIs, Cull et al. (2011) showed that profit seeking microfinance institutions maintain their profit rate by reducing or curtailing social outreach. It is also important to mention that those institutions who rarely focus on commercial gain and mostly focus on social outreach, sacrifice profits. This research is similar with the findings of Salim (2013) that, two of the largest MFIs in Bangladesh, namely Grameen Bank and BRAC, have sacrificed potential profit of 35.2% and 31.4% respectively in 2003 towards by providing greater social outreach.

Additionally, Serrano-Cinca and Gutiérrez-Nieto (2014) have found that, non –NGO and large MFIs (in terms of total assets) were statistically significant for drifting. Mersland and Strøm (2010) also documented that mature MFIs are mission-drifted when taking average loan size as a proxy of measurement.

Microfinance institutions are informal lending institutions and they are not well equipped with the latest Information and Communication Technology (ICT) and comprehensive products like the more formal financial institutions. Researchers have pointed out the importance of introducing ICT in their operation (Etim, 2013; Hishigsuren, 2006; Islam, Takanashi, & Natori, 2013; Kauffman & Riggins, 2012; Singh & Bhar, 2014). However, adoption of technology requires significant investment in capital, especially in the context of inherent manual labor oriented services. Those institutions who want to invest more capital need to be contrasted by labor usage in the form of resource constraints (Jin & Jin, 2014). Thus, MFIs might be interested to rely more on technology-driven services by incorporating less labor, but how they can bear such huge investment is an important issue to be addressed.

Capital simply refers to the availability of the financial resources. The form of capital in microfinance operations basically includes retained earnings, significant amount of donor's funds as well as funds from lenders. The primary assumption that this study considers is that the more financial capital (e.g. in a form of money) a firm has, the more likely they will invest in real capital (particularly referring to physical assets) for their production purpose.

Microfinance operations are generally labor intensive due to unique needs arising from the mission of providing financial services to poor communities (Ganga & Rizavi, 2006; Guntz, 2011; Nourse, 2001). Microfinance clients reside mostly in rural areas, have little or low income, low levels of education, and lack of technology usage, and the high cost of capital equipment is expected to drive the institutions to rely more on labor intensive services rather than capital intensive. However, as the sector matures and starts to provide more financial services (e.g. credit, savings, micro-insurance etc.) like formal financial institutions, investment in capital has started to rise, reducing operating costs for achieving financial sustainability (Ganga & Rizavi, 2006). Hence, we would like to examine the impact of such capital investment on financial sustainability versus social outreach performance.

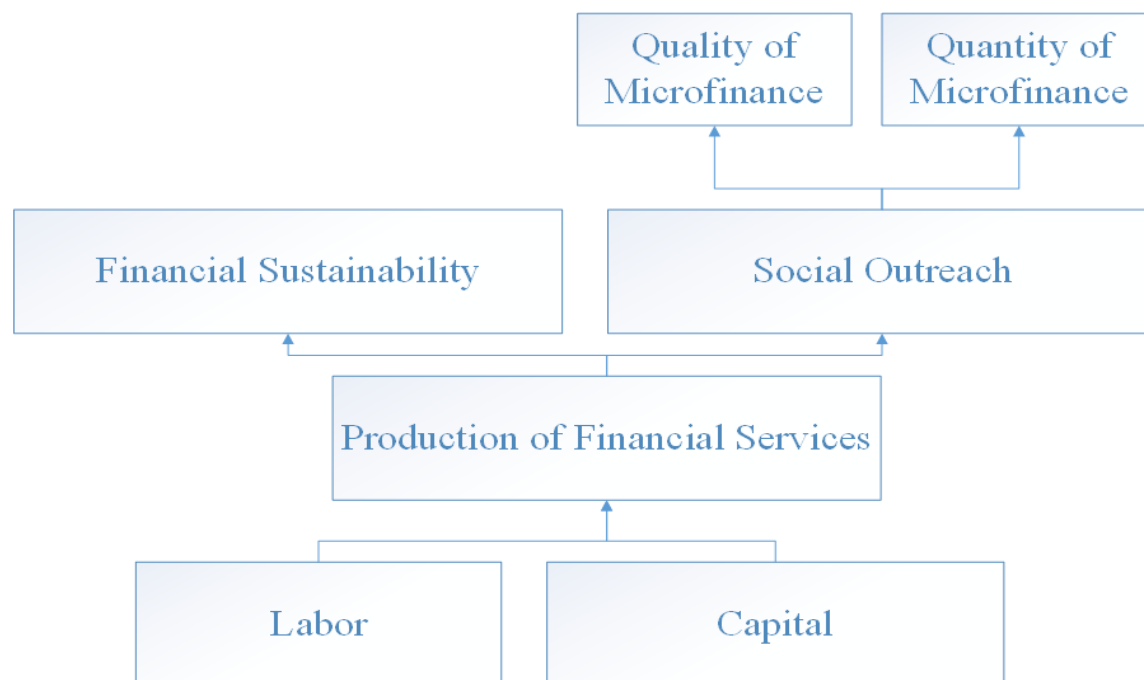
Developing countries are said to be labor abundant (Campbell & Ahmed, 2012) and the microfinance sector absorbs a substantial numbers of employees due to the low cost of labor. However, relying more on labor in aggregate terms is inefficient, increasing operational costs and thus the profitability of the institutions as well. The most important consideration of MFIs to be labor intensive is its selection procedures of borrowers. For example, the loan officers need to identify the borrowers and provide and collect the loan installments manually from the villages. This cannot be done by the technology, equipment or machinery. Although, some recent innovations such as mobile banking, online payment etc. have been incorporated in the banking sector, it is still rare in Bangladesh. Furthermore, how able the clients will be to use these systems is still unclear. After implementation of such costly projects, how effectively the borrowers will

use these services and whether they know how to operate such programs is a point to be pondered. However, usage of technology, such as computer and internet may be used in the administration process rather than manual calculation or estimation of loan records and other administrative jobs that would facilitate the employment of less administrative staff.

2.1 Resource Allocation: A Conceptual Framework

Resource allocation refers to the decisions made by organizations or firms where they maintain a balance between the combination of labor and capital (assuming these are the only two inputs of production) for producing goods and services and to maximize the goal of the entity. MFIs mainly have two inputs, namely labor and capital, to maximize both of their objectives. This study assumes that all MFIs are bound by resources constraints and a significant increase in one input must be accompanied by a respective reduction in another input. Based on the existing literature and a basic understanding of microfinance operation, the conceptual framework incorporating microfinance's dual objectives is shown below.

Figure1: Conceptual framework of microfinance operation.



Source: Author's own proposed framework

2.2. Hypothesis Development

Scholars agree on the point that lending to the poor is costly and there are several reasons for such a conclusion. (Hermes et al., 2011). Firstly, the poor require only a small amount of loan for their business and it is usually for the short term, in most cases one to three years. Managing small and short-term loans substantially increase the operating cost of the institutions (Ghosh, 2012; Hartarska, Shen, & Mersland, 2013). Secondly, the unique feature of this credit system is ‘collateral free’ loans, unlike the commercial banks where it is the prerequisite of financial transactions (Ghosh, 2012). Since it is collateral-free and associated risks of default are high, it pushes the cost upwards. Although group loan methodology have been addressed to minimize the risk, it can be harmful to social cohesion and there is a risk of discrimination among the various classes of the society (Ghosh, 2012).

To reduce the operating cost, some microfinance institutions are investing more on capital and relying less on labor. It is quite expected that most of the firms, whether formal or informal lending institutions, are rationale and, generate sufficient amount of revenue to cover their total cost. They would it is assumed, also do their best to allocate their resources (labor and capital) in such a way that will stimulate their performance and help to achieve their dual missions. Despite a growing trend of more capital and less labor, and the inherent labor intensive nature of microfinance operation, we derive a set of hypothesis that will be examined against the empirical estimation.

Hypothesis 1. Higher ‘capital-asset-ratio’ has a negative impact in microfinance revenue generation.

Hypothesis 2. More labor in the production increases the revenue generation of microfinance.

Secondly, as the quality, or depth, of social outreach requires the identification and selection of the deserving borrowers from the poor community, employees are more important to do this manual job than capital equipment.

Hypothesis 3: Higher ‘capital-asset -ratio’ has a negative impact on depth of outreach.

Hypothesis 4: More labor has a positive impact on depth of outreach.

In addition to providing credit, microfinance are extending their financial services to include savings, insurance and etc. for poor community. Marketing these products to the rural community requires more employees. Thus we hypothesize that,

Hypothesis 5: Higher ‘capital-asset-ratio’ deteriorates the quantity of microfinance social outreach.

Hypothesis 6: More labor involvement in microfinance operation enhances the quantity of microfinance social outreach.

Finally, adopting information and communication technology and increasing the capital investment proportionately increases the administrative cost of microfinance due to the high management cost and large depreciation cost. As a result of high labor demand, as well as unionization of the labor market, coincidental with other financial incentives to the employees, more labor engagement in operations positively increases the administrative cost as well. Thus we hypothesize that,

Hypothesis 7: Higher ‘capital-asset-ratio’ will increase the administrative cost.

Hypothesis 8: More labor engagement in microfinance production of financial services increases administrative costs.

3. Methodology

3.1. Selecting dependent and independent variable

Financial sustainability is one of the ‘buzzwords’ in the formal financial sector and the definition is not much different for the informal lending sector like microfinance. According to Ben Soltane (2012) when MFIs have a positive return on its asset and equity and operational self-sufficiency is over 100%, they are considered as financially sustainable. Furthermore, it also means that the institutions are said have positive net income and do not take government incentives or donations for subsidizing the operating expense or bad loans (Chaves & Gonzalez-Vega, 1996; Conning, 1999). According to Piot-Lepetit and Nzongang (2014), financial sustainability refers to efficient uses of resources to produce the services. Guntz (2011) also defined financial sustainability as covering total cost on adjusted basis and the ability to operate without ongoing subsidy in any forms (e.g. soft loans, government grants etc.). Although several indicators have been developed by scholars, there is no unanimity. Rather it is the subjective choice of the researchers that mostly capture the intended objective(s) of their studies. We have used the proposed framework developed by Yaron (1994) in both financial sustainability and social outreach. Financial revenue over asset is chosen as an indicator of financial sustainability, similar with other empirical studies, due to the importance of long term viability of the sector (Ben Soltane, 2014; Gutierrez-Nieto et al., 2007).

Social outreach, sometimes referred simply as ‘outreach’, is one of the broad areas that has been classified by the prominent researchers. Yaron (1994) has clearly stated the important indicators that can be used to estimate the social outreach, and this has been well used by other researchers (Ben Soltane, 2014; Gutierrez-Nieto et al., 2007). There are two dimension mostly looked by researchers, namely breadth of social outreach and depth of social outreach. We used both these terms as ‘quality’ and ‘quantity’ of microfinance in this study (Conning, 1999; Quayes, 2012).

Although there are no entirely appropriate indicators that can truly capture the depth (quality) of social outreach (Hoepner, Liu, & Spaggiari, 2011; Schreiner, 2001), use average loan size per borrowers as a proxy measure (Hisako, 2009; Quayes, 2012). Number of active clients, those who received credit services as well as other financial services such as savings, micro-insurance etc. are considered as the breadth of social outreach. In this study, we have used number of savers to capture the quantity of microfinance, in line with Yaron (1994) and Lafourcade, Isern, Mwangi, and Brown (2005). Additionally, to capture the associated cost in relation to investment in capital goods, administrative expense is considered as dependent variable as it changes in relation to the proportion of capital and labor.

In specifying the independent variables of this study, Cobb and Douglas (1928) production approach is used, where capital and labor are the two main inputs of production. Thus, we have used capital over asset and total number of employees as the two main inputs of microfinance. Capital over asset as a proxy of capital, means when an institution wants to increase investment on physical capital (machineries, land, equipment) they need to have sufficient funds. Thus, the higher the capital asset ratio the more funds are available to be invested as real capital.

Table 1: Definition of dependent and independent variable.

Variables	Definition	Expected Sign
<i>Fin.Rev</i>	Financial revenue over average total asset	N/A
<i>Avg.Loan</i>	Average loan outstanding per borrowers	N/A
<i>Num.Sav</i>	Number of total savers	N/A
<i>Adm.Exp</i>	Administrative expense over average total asset.	N/A
<i>CAR</i>	Total capital over total asset	-/+
<i>Num.Emp.</i>	Total number of employee	+

Source: Compilation from Microcredit Regulatory Authority, 2015.

3.2. Cobb-Douglas Production Function and Generalized Methods of Moment (GMM)

We classify MFIs as a producers of financial services as they provide credit, savings, micro-insurance etc. to their clients (Ben Soltane, 2014). A production function is a heuristic technique that shows how much output can be gained from a given level of inputs fixed to current level of technology (Miller, 2008). Similar to the Cobb and Douglas (1928) production models, where capital and labor are the two main component of production, our model is as follows:

$$\text{Output (financial services)} = f (\text{Employee, Capital})$$

$$\text{Output (Fin. Rev/Asset, Avg. Loan., and Num. Sav.)} = f (\text{Employee, Capital/asset})$$

Mathematically the trans-log production function derived the final econometric model as follows;

$$LnOutput_{it} = \partial_0 + \partial_1 LnOutput_{.it-1} + \partial_2 LnCAR_{it} + \partial_3 LnNum.Emp_{it} + \varepsilon_{it} \quad (1)$$

Where the subscript i and t denotes the specific microfinance institutions and time in years respectively. $LnOutput_{.it-1}$ is the first period lag of respective output. We are interested to estimate the coefficients of ∂_1 to ∂_3 and ε is the stochastic error term in the model. Furthermore, natural logarithm is used to improve the goodness of fit of the regression models and overcome simultaneity bias (De Bandt & Davis, 2000; Staikouras, Mamatzakis, & Koutsomanoli-Filippaki, 2008). Graphical pair-wise figures before and after the log transformation are provided in the Appendix A.

As we are dealing with dual objectives of microfinance the final econometric model for specific objectives are as follows

$$LnFin.Rev_{it} = \alpha_0 + \alpha_1 LnFin.Rev_{.it-1} + \alpha_2 LnCAR_{it} + \alpha_3 LnNum.Emp_{it} + \varepsilon_{it} \quad (2)$$

$$LnAvg.Loan_{it} = \beta_0 + \beta_1 LnAvg.loan_{.it-1} + \beta_2 LnCAR_{it} + \beta_3 LnNum.Emp_{it} + \varepsilon_{it} \quad (3)$$

$$LnNum.Sav_{it} = \gamma_0 + \gamma_1 LnNum.Sav_{.it-1} + \gamma_2 LnCAR_{it} + \gamma_3 LnNum.Emp_{it} + \varepsilon_{it} \quad (4)$$

$$LnAd.Exp_{it} = \varphi_0 + \varphi_1 LnAdm.Exp_{.it-1} + \varphi_2 LnCAR_{it} + \varphi_3 LnNum.Emp_{it} + \varepsilon_{it} \quad (5)$$

The equation 2, 3, 4 and 5 are the main focus of this study as we are interested to establish the relationship between capital and employment with financial, depth and breadth of social outreach and administrative expense respectively.

Estimation with lagged dependent variables are known to suffer from endogeneity bias since the lagged dependent variable is systematically correlated with the error term. We have used GMM estimator proposed by Arellano and Bond (1991) that eliminates any endogeneity may be due to the correlation of country specific, time-invariant factors and the right hand side repressors. Lagged levels of the dependent variable, predetermined variables and endogenous variables are used to form GMM-type instruments. First differences of the strictly exogenous variables are used as standard instruments and uncorrelated with the fixed effect, system GMM (SGMM) allows more instrument that enhance the efficiency (Roodman, 2006). As endogeneity may effect robustness of the estimation, this study uses SGMM.

As for the diagnostics tests, Sargen and Hansen test are reported in the empirical findings. Both the Sargen and Hansen test of over-identification examines the appropriateness of the instruments used. Arellano-Bond AR (1) and AR (2) tests the existence of the first and second-order serial correlation in the first-differenced residuals.

3.3. Data

The data for this study was obtained from the Microcredit Regulatory Authority (www.mra.gov.bd) database that governs and monitors the operation of MFIs in Bangladesh. This organization was established in 2006 in line with government policy to increase transparency of MFIs. The database was released in 2011 and was open to the public. The study covered the period from 2007 to 2012 and 176 MFIs were selected, based on the total sample of 643. These MFIs comprise non-governmental organizations (NGOs), credit unions, cooperatives, non-banking financial institutions (NBFIs), special credit program of the ministry, state-owned commercial banks and private commercial banks. Outliers were detected and appropriate measurements were taken into consideration.

4. Discussion on Empirical Findings

Table 2 reports the basic statistics of the variables considered in this study. The mean value of financial revenue over asset was 0.971, showing that the MFIs are generating revenue efficiently from their total assets. The maximum was recorded at 2.96, simply referring to the ability of generating taka 2.96 of revenue for every taka 1 of asset. The highest number of savers was 780232, showing the sample also included large MFIs. Although the mean average loan size was small (approx. US\$88) with the highest recorded at 21675.220 (approx. US\$277). As it is costly to lend to the poor, the average administrative cost was over 12% of average total asset, while it soared up to taka 1.03 for every one taka of asset held, which is comparatively high. The capital asset ratio was moderate at a mean value of 0.302. In terms of number of employees, there were some MFIs that were as low as 3 employees to almost three thousand.

Table 2: Descriptive statistics (in Taka) (2007-2012).

Variable	Obs.	Mean	Std. Dev.	Min	Max
Fin.Rev	1036	0.971	0.612	0.040	2.96
Num.Sav.	1050	36430	72809	703	780232
Avg.Loan.(Taka) ^a	1054	6927.534	3007.141	1412.310	21675.220
Adm.Exp.	1034	0.123	0.078	0.010	1.030
CAR	1040	0.302	0.185	0.010	1.300
Num.Emp.	1049	213.001	369.10	3	2923

Source: Author's own estimation. ^a On average, US\$1= Taka 78.10 in 2013 according to World Bank.

This study was mainly concerned with estimating how the CAR and employee related to microfinance financial and social outreach performance. Table 3 reports the empirical estimation of GMM in relation to the two inputs for attaining both financial and social outreach sustainability. Considering financial revenue, quality and quantity of social outreach (Model-1 to Model 3), we have found that capital asset ratio are negative while number of employees was positively related in all conventional levels of significance (1%, 5%, 10%). In the operation of microfinance, liquidity is more important than fixed asset such as, land, machinery etc. As the sector is informal and mostly deals with rural poor communities, comprehensive machineries, technological equipment (ATM, internet etc.) cost much investment in fixed asset and leave less funds for loan

disbursement. It is reasonable to assume that MFIs have constraints of financing and there is exists a trade-off between investing in fixed asset and available funds for loans in this context. As the main sources of MFIs earning is coming from interest and fees income from their outstanding loans, a 1% increase in capital asset ratio will reduce the financial revenue earnings by 0.50% assuming other things remain same or *ceteris paribus*. Furthermore, higher capital asset ratio also significantly has a negative impact on quality and quantity of microfinance. From the empirical estimation we have observed that 1% increase in capital asset ratio is likely to reduce the quantity and quality of microfinance by 0.13% and 0.07% respectively. These percentage-changes are less than the percentage-change in financial revenue.

Table 3: Results of GMM estimation

Independent variable	Model-1 (Fin.Rev)	Model-2 (Num.Sav)	Model -3 (Avg.Loan)	Model 4 (Adm.Exp)
Lag-1 (Dependent Variable)	0.155*** ^a (0.059)	0.113** (0.041)	0.925*** (0.032)	-0.031 (0.700)
CAR	-0.501*** (0.086)	-0.134** (0.051)	-0.071** (0.034)	0.258** (0.094)
Num.Emp.	0.125*** (0.038)	0.888*** (0.025)	0.050*** (0.014)	0.013 (0.036)
Constant	-1.410 *** (0.195)	4.300*** (0.386)	0.451*** (0.293)	-1.959*** (0.253)
Number of Instruments	63	62	63	62
AR(1)(p) ^b	-2.60 (0.000)	-3.74 (0.000)	-3.86 (0.000)	-1.31 (0.000)
AR(2)(p) ^b	1.22 (0.221)	-0.54 (0.591)	0.06 (0.955)	1.41 (0.157)
Sargan Test(p) ^b	629.21 (0.000)	529.48 (0.000)	814.45 (0.000)	749.49 (0.000)
Hansen Test(p) ^b	91.64 (0.004)	90.46 (0.004)	99.91 (0.001)	88.14 (0.007)
Number of Observation	839	862	866	844

Source: Author's own estimation. Values in parenthesis show the standard error.

a. *, **, *** indicates statistical significance at 10%, 5% and 1% respectively.

b. *P*-values are in parenthesis.

While employees (particularly loan officers), in charge of identifying potential clients are significant, both dimensions of social outreach and financial sustainability met the expected hypothesis. Revenue is generated from loans given to the borrowers. The more loans that are given, the more revenue generation is assured. Additionally, most of the time institutions require the loan officers to achieve certain levels of loan disbursement to achieve the institutional target. For example, 1% increase in labor will generate 0.13% more revenue over asset for the institutions. Furthermore, the quality and quantity is enhanced by 0.89% and 0.05% for every additional 1 % increase in labor engagement in microfinance operation. Labor has greater contribution in quantity (number of savers) of microfinance operation as it also requires the loan officers to convince clients to obtain extended financial services such as savings. Employees also receive training from donors, development institutions and other international agencies that improves their performance in providing the services to the poor community, coincidental with better institutional performance. Sila (2014) has emphasized the positive impact of employee training in microfinance operation

and service delivery. Overall, the MFIs also follow the strict enforcement in their operation by following ‘carrot and stick’ approach. If an employee shows good performance in terms of repayment collection, achieving borrowers target as well as clients satisfaction in terms of their services, they are rewarded respectively (such as bonus, promotion etc.) While failing to fulfill such criteria’s, MFIs also punish them by fine, no promotion etc. Thus, our expectation are quite likely that, these kind of rules and regulations impose towards an employee not only financially benefits the MFIs, but also towards achieving their social mission.

As we also intended to investigate how the capital asset ratio and employee impact administrative expense, we have observed the expected signs for both. The number of employees was not significant in any conventional level, while capital asset ratio was significant in 5% and 10% level. Higher capital asset ratio is estimated to increase the administrative cost by 0.26% for every additional percentage increase in the former. It shows that, when capital is invested in terms of physical assets, equipment’s and etc. it requires a substantial amount of administrative costs for managing that capital. As such, it increases the cost.

Regarding the GMM estimation, our models are free of serious autocorrelation based on the AR(2) values. AR(2) values shows that there is no autocorrelation in the second order of residuals, while there is existence in first order AR(1) residual, which is expected due to the built in mathematical program. The main concern of this model is its robustness. Based on the Sargen test for each of the model, we have observed that they are not robust but not weakened by many instruments. Furthermore, although the Hansen test shows that these models are robust, that robustness was weakened by the many instruments involved in our empirical estimation. These are the common phenomena in empirical estimation in linear GMM and nonlinear GMM (Stock, Wright, & Yogo, 2002).

4.1. Output Elasticity and Returns to Scale in Microfinance

From the table 3, we can derive the elasticity of labor and capital as we have log transformed both of the values. The coefficients of each of the variables shows the percentage changes in one input will lead to percentage increase/decrease to output, assuming *ceteris paribus* conditions held. From the empirical findings we observed that partial output elasticity of labor is less than 1 for all the 4 models, meaning that one percent increase in labor as an input will produce output less than one percent, assuming other things remain same. Quantity or breadth of outreach was highly sensitive to the labor changes, and for every additional percentage increase it will increase the number of savers to almost 0.89% as estimated, followed by financial sustainability and depth (quality) of outreach. On the other hand, 1% increase in CAR is estimated to reduce the financial revenue, number of savers and average loan size by 0.50%, 0.13% and 0.07% respectively. However, the estimated increase in administrative cost will be 0.26% for one percent unit increase in CAR.

The sector also experienced decreasing returns to scale as we add up the coefficients of capital and employees together (it is less than 1). It shows that the relative increase in outputs are less than a proportional increase in inputs (combining both CAR and employee).

Conclusion

This study endeavors to establish the relationship of CAR and number of employees in microfinance quality, quantity and financial sustainability. By using system GMM with a panel data of 176 MFIs between 2007 and 2012 in Bangladesh, this study showed interesting results. We have observed that higher capital asset ratio is estimated to reduce the revenue generation ability of microfinance institutions reducing financial sustainability. Furthermore, more capital intensity in relation to total asset also adversely affect quality and quantity of social outreach with different magnitudes and are statistically significant. As theoretically expected, the result also empirically showed that higher employee involvement in microfinance operation will enhance its financial sustainability and stimulate both quality and quantity of microfinance. Thus, microfinance should remain labor intensive for creating a better impact to society.

Additionally, relying more on capital goods and equipment, financed by higher capital ratio, also tends to increase the administrative cost of the organization. However, we failed to statistically prove that more employment will increase the cost in microfinance operation. We have statistically proved all of our hypothesis except the last one, which is the expected positive relationship between administrative expense and number of employees in microfinance operation.

In line with our empirical findings, this study concludes that higher capital asset ratio or the tendency of MFIs administration to invest much in capital goods will not bring expected benefit to their financial sustainability and in fact has an adverse impact.. Thus, it is most likely that, those MFIs investing much on capital are less likely to contribute more on social outreach, particularly quality and quantity. However, we could not find that more capital investment enhances financial sustainability. Rather it deteriorates sustainability. Management should take precautionary measurements in this regard before they increase their capital asset ratio and use these capital for investing in capital goods and equipment.

This study has some limitations. Proxy used in this study may partially reflect the intended content of our analysis, while further research may look into the appropriate measurements of respective variables. As we tried to address the impact of CAR and number of employees in microfinance's 'dual mission', which combination of both are complementary for financial sustainability and social outreach remains an important issue to explore in the future.

References

- Adhikary, S., & Papachristou, G. (2014). Is there a trade-off between financial performance and outreach in south asian Microfinance institutions? *The Journal of Developing Areas*, 48(4), 381-402.
- Annim, S. K. (2012). Microfinance efficiency: trade-offs and complementarities between the objectives of microfinance institutions and their performance perspectives. *European Journal of Development Research*, 24(5), 788-807.
- Arellano, M., & Bond, S. (1991). Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. *The review of economic studies*, 58(2), 277-297.

- Armendáriz, B., & Szafarz, A. (2011). On mission drift in microfinance institutions. *The handbook of microfinance*, 341, 366.
- Augsburg, B., & Fouillet, C. (2013). Profit empowerment: The microfinance institution's mission drift. *The Credibility of Microcredit: Studies of Impact and Performance*, 199.
- Ben Soltane, B. (2012). Social and financial performance of microfinance institutions: Is there a trade-off? *Journal of Economics and International Finance*, 4(4), 92-100.
- Ben Soltane, B. (2014). Total factor productivity change of MENA microfinance institutions: A Malmquist productivity index approach. *Economic Modelling*, 39, 182-189.
- Campbell, D., & Ahmed, I. (2012). The labour market in developing countries. *Perspectives on Labour Economics for Development (ILO, 2013)*.
- Chaves, R. A., & Gonzalez-Vega, C. (1996). The design of successful rural financial intermediaries: Evidence from Indonesia. *World Development*, 24(1), 65-78.
- Cobb, C. W., & Douglas, P. H. (1928). A theory of production. *The American Economic Review*, 139-165.
- Conning, J. (1999). Outreach, sustainability and leverage in monitored and peer-monitored lending. *Journal of Development Economics*, 60(1), 51-77.
- Copestake, J. (2007). Mainstreaming microfinance: social performance management or mission drift? *World Development*, 35(10), 1721-1738.
- Cull, R., Demirgüç-Kunt, A., & Morduch, J. (2011). Does regulatory supervision curtail microfinance profitability and outreach? *World Development*, 39(6), 949-965.
- De Bandt, O., & Davis, E. P. (2000). Competition, contestability and market structure in European banking sectors on the eve of EMU. *Journal of Banking & Finance*, 24(6), 1045-1066.
- Etim, A. S. (2013). Mobile technology adoption for microfinance delivery in Sub-Saharan Africa. *Research in Business & Economics Journal*, 7.
- Ganga, G., & Rizavi, S. (2006). *Microfinance Sector and Poverty Alleviation in Guyana: An Overview*. Paper presented at the XXXVII Annual Monetary Studies Conference, Barbados.
- Ghosh, J. (2012). Microfinance and the Challenge of Financial Inclusion for Development. *Ensayos Económicos*, 1(67), 7-34.
- Guntz, S. (2011). Sustainability and profitability of microfinance institutions. *CAIFD-Center for Applied International Finance and Development, Research paper*, 4, 2011.
- Gutierrez-Nieto, B., Serrano-Cinca, C., & Mar Molinero, C. (2007). Microfinance institutions and efficiency. *Omega*, 35(2), 131-142.
- Hartarska, V., Shen, X., & Mersland, R. (2013). Scale economies and input price elasticities in microfinance institutions. *Journal of Banking & Finance*, 37(1), 118-131.
- Hermes, N., Lensink, R., & Meesters, A. (2011). Outreach and efficiency of microfinance institutions. *World Development*, 39(6), 938-948.
- Hisako, K. (2009). Competition and wide outreach of microfinance institutions. *Economics Bulletin*, 29(4), 2628-2639.
- Hishigsuren, G. (2006). *Information and communication technology and microfinance: Options for Mongolia*: Asian Development Bank Institute.
- Hoepner, A. G., Liu, H., & Spaggiari, L. (2011). The Outreach Measurement Debate in Microfinance: Does Average Loan Size Relate to Client Poverty? Available at SSRN 1956569.
- Islam, M. T., Takanashi, C., & Natori, T. (2013). Big Challenges by a Small Microfinance Institution-A Case Study of SafeSave Bangladesh from the Customer Satisfaction and ICT Introduction Perspective. *International journal of Business and Management*, 8(14), p23.
- Jin, L., & Jin, J. C. (2014). Internet Education and Economic Growth: Evidence from Cross-Country Regressions. *Economies*, 2(1), 78-94.
- Kauffman, R. J., & Riggins, F. J. (2012). Information and communication technology and the sustainability of microfinance. *Electronic Commerce Research and Applications*, 11(5), 450-468.
- Lafourcade, A.-L., Isern, J., Mwangi, P., & Brown, M. (2005). Overview of the outreach and financial performance of microfinance institutions in Africa. *Microfinance Information eXchange*, Washington, DC. http://www.mixmarket.org/medialibrary/mixmarket/Africa_Data_Study.pdf.

- Louis, P., Seret, A., & Baesens, B. (2013). Financial efficiency and social impact of microfinance institutions using self-organizing maps. *World Development*, 46, 197-210.
- Mersland, R., & Strøm, R. Ø. (2010). Microfinance mission drift? *World Development*, 38(1), 28-36.
- Miller, E. (2008). *An assessment of CES and Cobb-Douglas production functions*: Congressional Budget Office.
- Morduch, J. (2000). The microfinance schism. *World Development*, 28(4), 617-629.
- Nourse, T. H. (2001). The missing parts of microfinance: Services for consumption and insurance. *SAIS Review*, 21(1), 61-69.
- Piot-Lepetit, I., & Nzongang, J. (2014). Financial sustainability and poverty outreach within a network of village banks in Cameroon: A multi-DEA approach. *European journal of operational research*, 234(1), 319-330.
- Quayes, S. (2012). Depth of outreach and financial sustainability of microfinance institutions. *Applied Economics*, 44(26), 3421-3433.
- Roodman, D. (2006). An introduction to difference and system gmm in stata. *Center for Global Development Working Paper Series*.
- Salim, M. M. (2013). Revealed Objective Functions of Microfinance Institutions: Evidence from Bangladesh. *Journal of Development Economics*, 104, 34-55.
- Schreiner, M. (2001). Seven aspects of loan size. *Journal of Microfinance/ESR Review*, 3(2), 27-47.
- Serrano-Cinca, C., & Gutiérrez-Nieto, B. (2014). Microfinance, the long tail and mission drift. *International Business Review*, 23(1), 181-194.
- Sila, A. K. (2014). Relationship between training and performance: A case study of Kenyan women finance trust eastern nyanza region, Kenya. *European Journal of Business and Social Science*, 3(1), 95-117.
- Singh, R., & Bhar, C. (2014). The Growth of Microfinance and the Role of DSS in Reducing Its Volatility. *IUP Journal of Management Research*, 13(4).
- Staikouras, C., Mamatzakis, E., & Koutsomanoli-Filippaki, A. (2008). An empirical investigation of operating performance in the new European banking landscape. *Global Finance Journal*, 19(1), 32-45.
- Stock, J. H., Wright, J. H., & Yogo, M. (2002). A survey of weak instruments and weak identification in generalized method of moments. *Journal of Business & Economic Statistics*, 20(4).
- Tchuigoua, H. T. (2014). Institutional framework and capital structure of microfinance institutions. *Journal of Business Research*, 67(10), 2185-2197.
- Woller, G. (2007). Trade-offs between social and financial performance. *Journal of Microfinance/ESR Review*, 9(2), 14-19.
- Woller, G., & Schreiner, M. (2002). Poverty lending, financial self-sufficiency, and the six aspects of outreach. *Disc. Paper, Ohio*.
- Yaron, J. (1994). What makes rural finance institutions successful? *The World Bank Research Observer*, 9(1), 49-70.
- Zerai, B., & Rani, L. (2012). Is there a tradeoff between outreach and sustainability of micro finance institutions? evidence from indian microfinance institutions (MFIs). *Research Journal of Finance and Accounting*, 2(11), 32-41.